

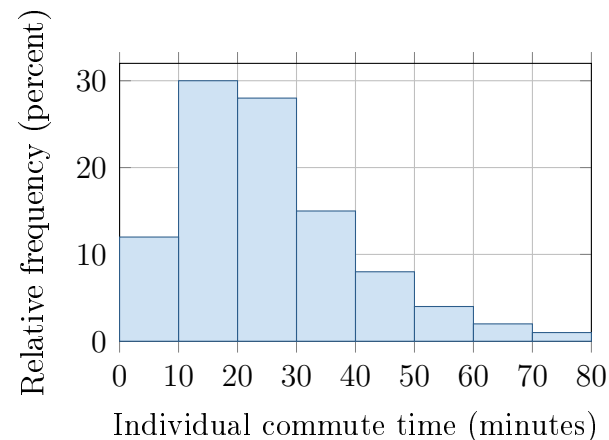
# When Is the Mean Nearly Normal?

Unit 2 · Topic 2.12 · TODAY'S PROBLEM

Individual commute times in a very large city are strongly right-skewed, with population mean  $\mu = 25$  minutes and standard deviation  $\sigma = 15$  minutes. Consider independent random samples.

**Rae:** “Any average is approximately normal, even for  $n = 4$ .”

**Sol:** “With this long right tail,  $n = 4$  may be too small;  $n = 40$  is more defensible.”



## FIRST TAKE (5 min, on your sheet)

Which would you expect to smooth out the long commute times more: averaging 4 people or 40? Give one reason.

- DOK 1** (a) Identify the population shape and state the population mean and standard deviation with units.
- DOK 2** (b) For  $n = 4$  and  $n = 40$ , calculate the mean and standard deviation of the sampling distribution of  $\bar{x}$ .
- DOK 3** (c) In 3–4 sentences, decide whose claim is better supported. Use the population shape, independence, and the two sample sizes. Explain which distribution the CLT describes and why using a normal curve for averages of 4 could be unreliable.

**Optional review:** [follow-along](#) (scan the code)

Work (a)–(c) from the rules on your sheet. Turn in your sheet.

